F · Legs, feet, rails and cabinets

Tops face down on a blanket: six legs, eight feet (gauged before the plates go on) and the two rails. Then the cabinets are set by the walls. Rings come in chapter G.

No.	Part	No.	Hardware	Qty
P01 **P02**	Top A and top B, EKBACKEN 249 × 65 cm, 28 mm thick	**H01**	M4 x 16 mm machine screw, IKEA 108443 (with the OLOV legs)	30
P03 **P04**	ALEX 1, drawer unit; ALEX 2, storage unit with door (36 × 58 × 70 cm)	**H02**	#8 x 5/8 in pan head screw (feet plates, 4 each)	32
P05	OLOV leg x 6, 105 mm plate	**H06**	#6 x 5/8 in flat head (rings, 3 each)	24
P06 **P07**	Rail A and rail B, steel angle 213 cm, drilled	**H03**	#10 x 1 in pan head, Robertson (rails; 17 now, 2 in chapter H)	19
P09 **P10**	Round adjustable foot 4 in x 8; printed locating ring x 8 (chapter G)	**P18** printed tools: leg jig, feet corner jig, edge T-square, ALEX corner template, drill guide (tower + pucks), foot height gauge, wall gauge		

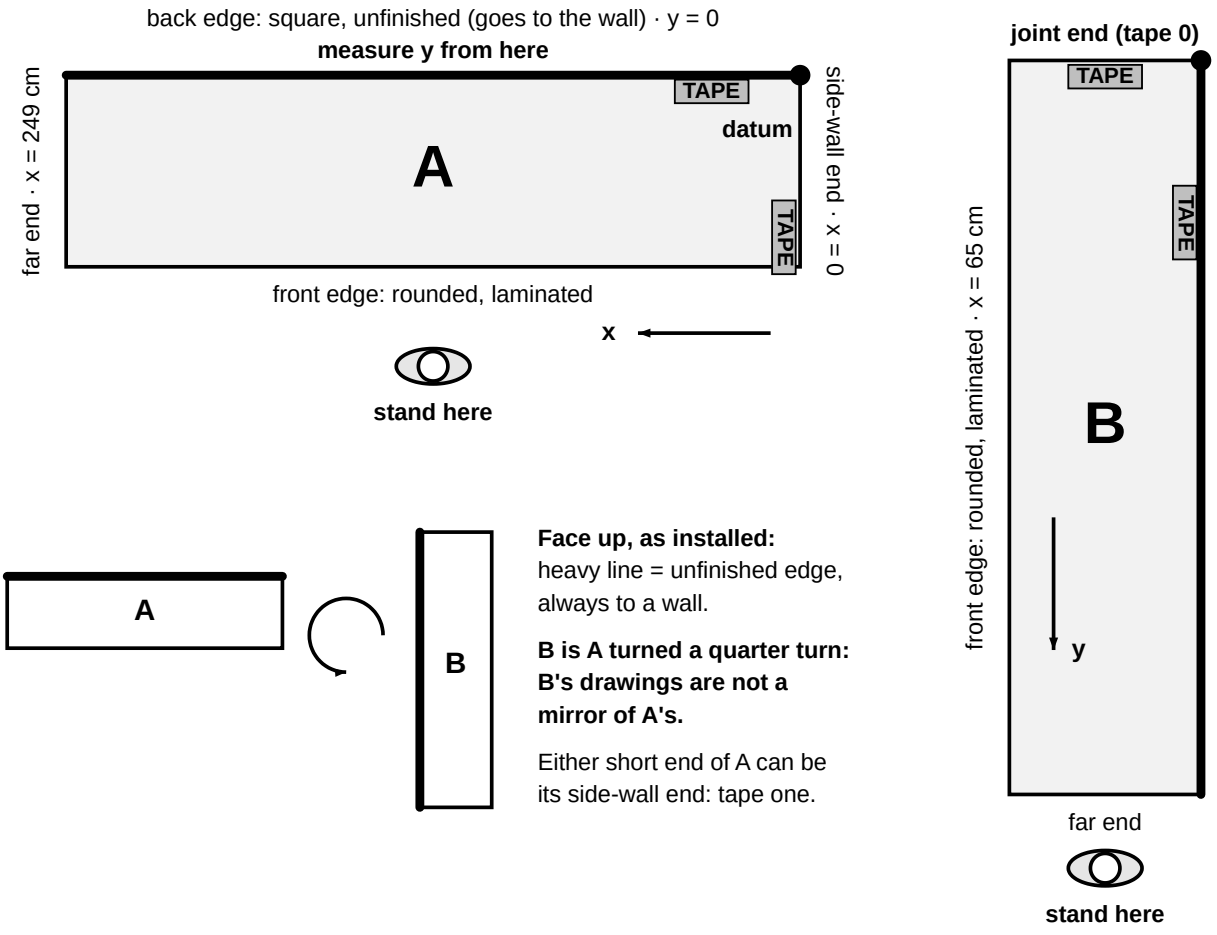


Fig. F-1 How each top lies while you mark it: face down, laminate on the blanket. Heavy line: the square, unfinished long edge, which always goes to a wall (y = 0 on A, x = 0 on B). Tape the unfinished edge and the end you measure from, and stand where the person symbol is; the drawings in this chapter then match what you see, with x growing to the left. Datum: top A's back-left corner (0, 0). The arrow turns A anticlockwise into B.

Decide the desk height first (the legs are pre-set in step 1): built at 83.0 cm as planned; optional printed ALEX feet, 2, 4 or 6 cm tall, raise everything by their height. SOURCE-DEFINED

ALEX feet	Underside of the tops (cm)	Top surface (cm)	Every OLOV set to (cm)	Rail lower edge (cm)
none (as planned)	80.2	83.0	80.2	75.1
2 cm	82.2	85.0	82.2	77.1
4 cm	84.2	87.0	84.2	79.1
6 cm	86.2	89.0	86.2 (OLOV max 90)	81.1

Contents

F · Legs, feet, rails and cabinets 1

F · Underside support layout

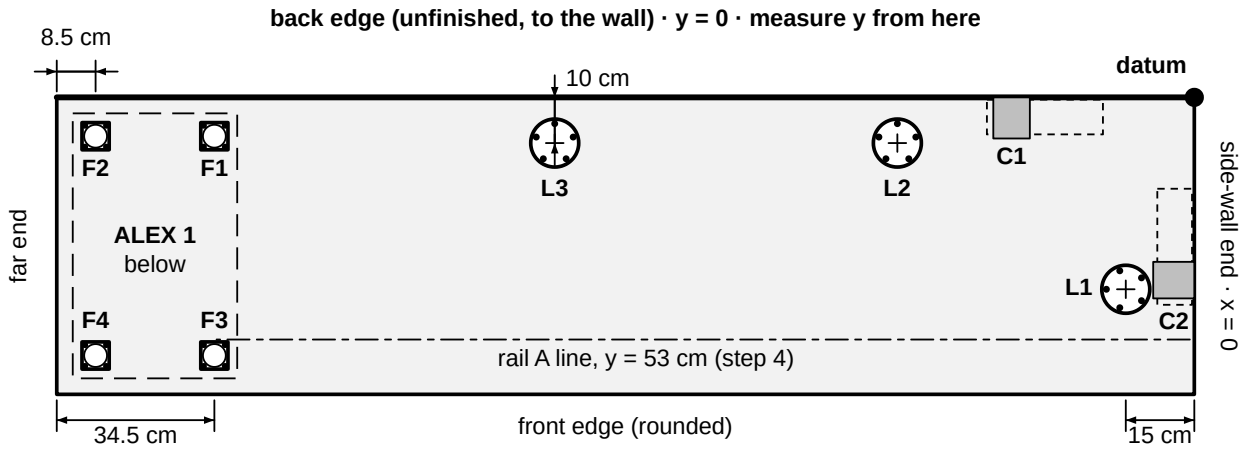


Fig. F-2 Top A face down. Stand at the front edge with the back edge away from you: the side-wall end is on your right and x grows to the left from the side-wall end. Circles: OLOV plates with their five holes, one pointing at the wall (L1's pattern is turned). Squares: feet plates. Grey: clamp zones C1, C2, keep clear; thin dashed: the clamp pad outlines; dash-dot: rail A's line. Datum (0, 0) at the top-right corner of this view. Drawing labels in cm.

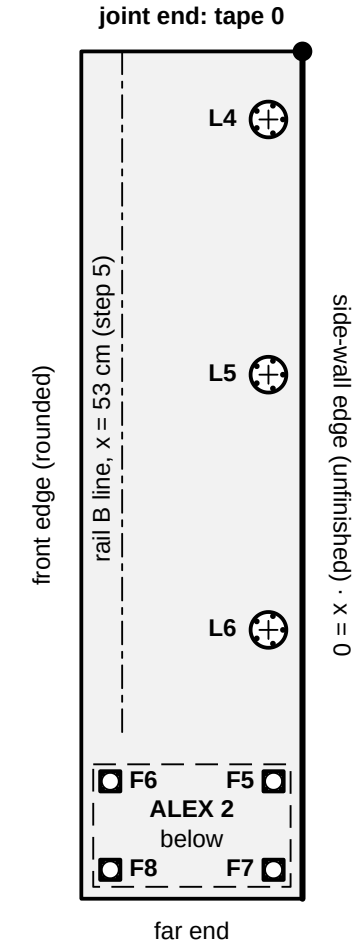


Fig. F-3 Top B face down. Stand at B's far end with the joint end away from you: the side-wall edge is on your right and x grows to the left. Legs: tape hooked on the joint end; feet: tape hooked on the far end.

Mark	x	y	Tape reading
L1	15	42	42 from A's back edge
L2	65	10	65 from the side-wall end
L3	140	10	140 from the side-wall end
L4	10	85	20 from B's joint end
L5	10	160	95 from the joint end
L6	10	235	170 from the joint end
F1	214.5	8.5	34.5 from A's far end
F2	240.5	8.5	corner jig, far back corner
F3	214.5	56.5	34.5 from A's far end
F4	240.5	56.5	corner jig, far front corner
F5	8.5	279.5	34.5 from B's far end
F6	56.5	279.5	34.5 from B's far end
F7	8.5	305.5	corner jig, far wall corner
F8	56.5	305.5	corner jig, far front corner

x, y and tape readings in cm; x, y from the datum **SOURCE-DEFINED**. Feet are 5 cm in from each ALEX corner, which is 8.5 cm in from the top's edges. F1 F3 F5 F6 are always measured from the far end, the same end the cabinet is set from, never from the other end.

Leg plates are ⌀105 mm and feet plates 59.5 mm square. L1's plate ends short of rail A and clear of the clamp 2 jaw; F3's plate stays 5 mm clear of rail A; F6 sits 5 mm outside rail B's line, but rail B stops 13.5 cm short of it.

> Nothing here goes in the clamp zones C1 (36–44 cm from the side-wall end, the 9 cm next to the back edge) or C2 (36–44 cm from the back edge, the 9 cm next to the side-wall end).

F · Step 1: legs

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> Calipers on both tops before the first hole: expect 28 mm. If a top reads 27 mm, use screws 1 mm shorter. No screw may go deeper than 21 mm into a top.

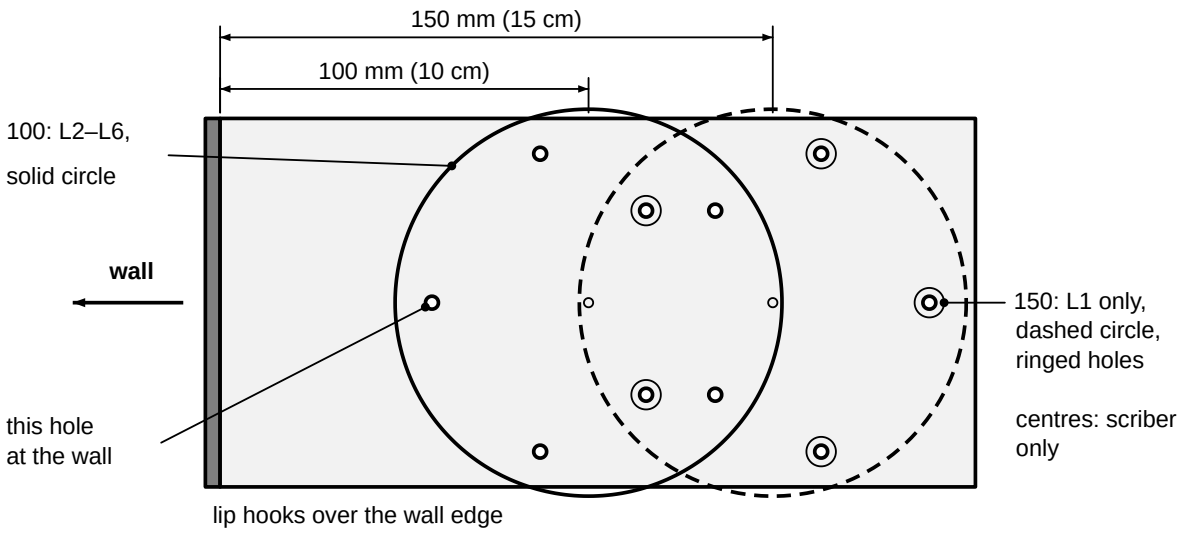


Fig. F-4 Leg jig (P18), seen from above in use, lip hooked down over the wall edge. The 100 pattern (10 cm from the edge; solid plate circle, plain holes) is for L2 to L6 and puts one hole at the wall; the 150 pattern (15 cm; dashed circle, ringed holes) is for L1 only.

Leg	Top	Hook the lip over	Tape station (cm)	Pattern
L1	A	side-wall end (x = 0)	42 from the back edge	150, ringed holes
L2 L3	A	back edge (y = 0)	65 (L2) and 140 (L3) from the side-wall end	100
L4 L5 L6	B	side-wall edge (x = 0)	20 (L4), 95 (L5), 170 (L6) from the joint end	100

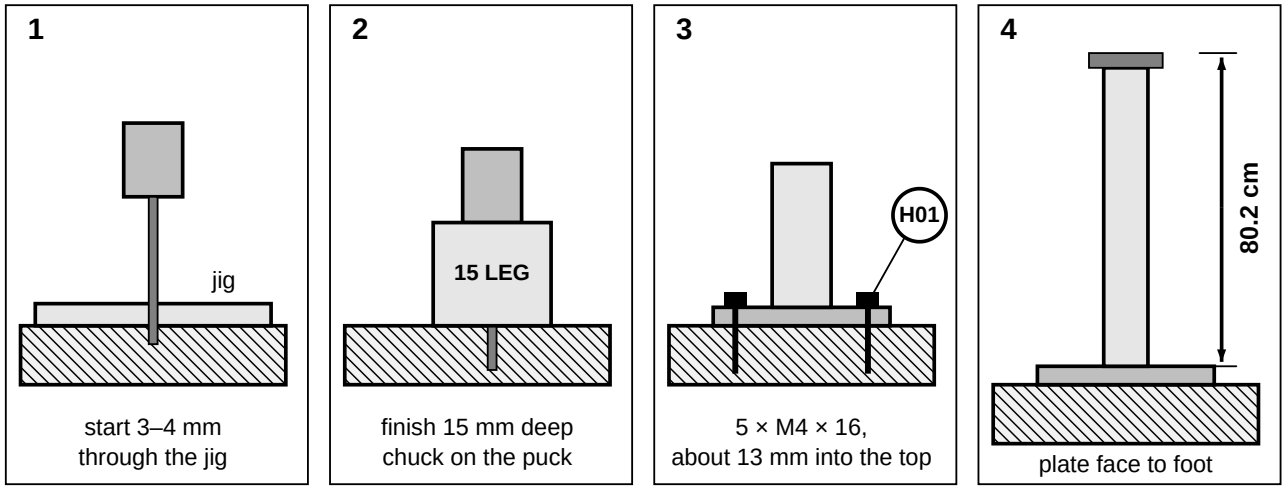


Fig. F-5 One leg, four operations (steps 1a–1d below). The drill guide's SET tower sets the 1/8 in bit first.

1

a Jig on the station, start all five pilots 3–4 mm through its holes with the 1/8 in bit. **b** Lift the jig; finish each pilot 15 mm deep with the **15 LEG** puck. **c** Screw the OLOV plate on with five **H01** M4 × 16 screws (about 13 mm into the top). **d** Pre-set the leg to **80.2 cm** from plate face to foot (or the table's number at the start of this chapter if you fit ALEX feet). Six legs, 30 screws.

OLOV plate ⌀105 mm, five ⌀5.5 mm holes on an ⌀85 mm circle, neighbouring holes 50 mm apart; leg range 60–90 cm. VERIFIED
105 mm plate measured 22 Oct 2026; hole pattern from a shared OLOV template, printed and test-fitted.

F · Step 2: wind each foot to 102 mm, loose

ASSUMPTION / FIELD CHECK ASSUMPTION / FIELD CHECK
Measure your ALEX units first, with and without their stock feet. The drawings take an ALEX as 70 cm tall, which is what makes the feet 102 mm between the ALEX top and the underside at 80.2 cm.

2

Hold each foot loose in your hand, plate and all, before anything is screwed. Wind its pad out until the printed foot height gauge (102.0 mm inside) just slides over it from the side with no rock. All eight then match without a ruler. Do this now: the gauge's C sits against the plate face, so it cannot be used once the plate is screwed to the top.

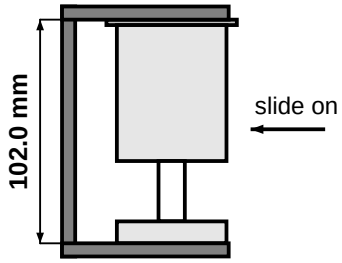


Fig. F-6 Foot height gauge on a loose foot. The C bears on the plate face and on the pad.

ASSUMPTION / FIELD CHECK ASSUMPTION / FIELD CHECK
The feet's load rating is unknown; the design wants 50 kg or more each.

F · Step 3: feet plates

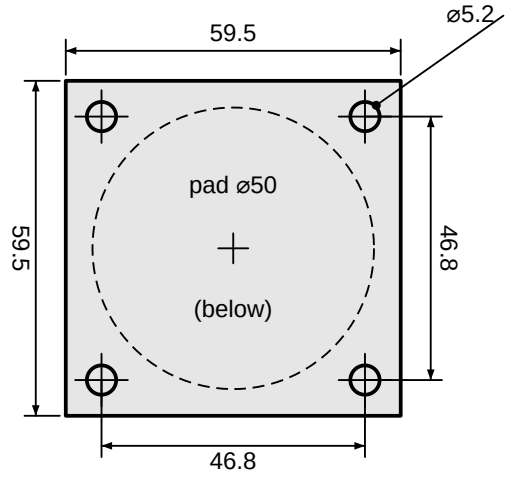
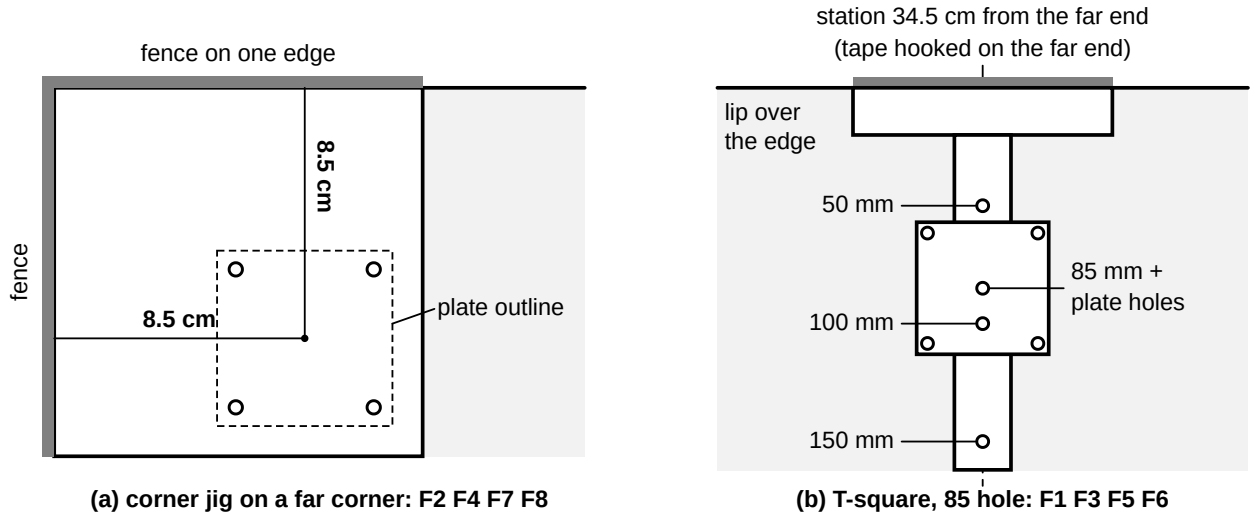


Fig. F-7 Feet plate (P09) from the screw side, mm. **VERIFIED** on the purchased plate, 23 Oct 2026: 41.6 mm between two neighbouring holes (inside jaws), 52.0 mm across them (outside jaws).

Feet	Tool	Placement
F2 **F4**	feet corner jig	top A's two far corners, fences on both edges
F1 **F3**	T-square, 85 hole	lip over A's back edge (F1) or front edge (F3), station 34.5 cm from A's far end (tape hooked on the far end)
F7 **F8**	feet corner jig	top B's two far corners
F5 **F6**	T-square, 85 hole	lip over B's side-wall edge (F5) or front edge (F6), station 34.5 cm from B's far end (tape hooked on the far end)



(a) corner jig on a far corner: F2 F4 F7 F8

(b) T-square, 85 hole: F1 F3 F5 F6

Fig. F-8 Both tools put the foot's four plate holes (1/8 in, on the 46.8 mm square) square to the edges in one placement, seen face down. Centre holes are for a scriber only. T-square offsets are in mm, as engraved on it.

3

Start the four pilots 3–4 mm through the tool's plate holes with the 1/8 in bit, lift the tool, finish 15 mm deep with the **15 LEG** puck. Screw the plate of each gauged foot on with four **H02** #8 × 5/8 pan heads (14 mm into the top), pad pointing away from the top. Eight plates, 32 screws.

F · Steps 4 and 5: rails to the tops

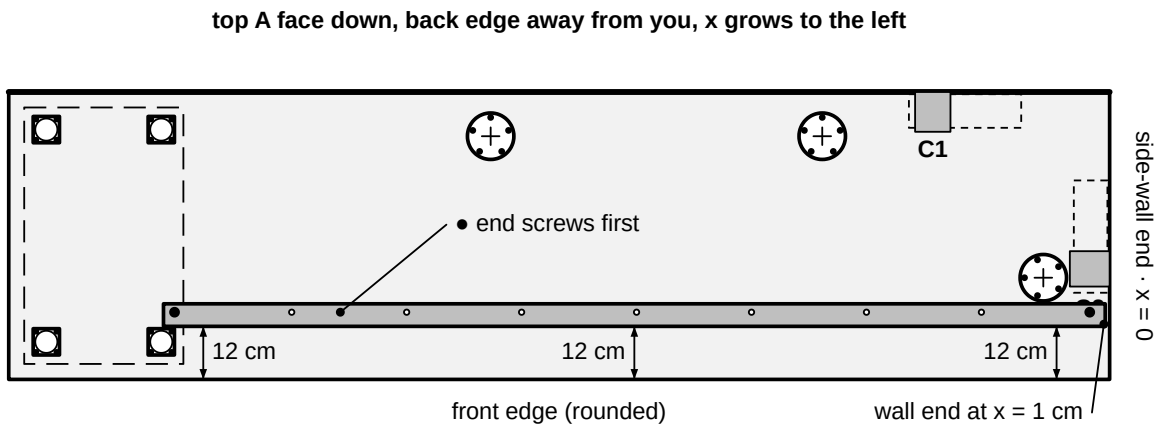
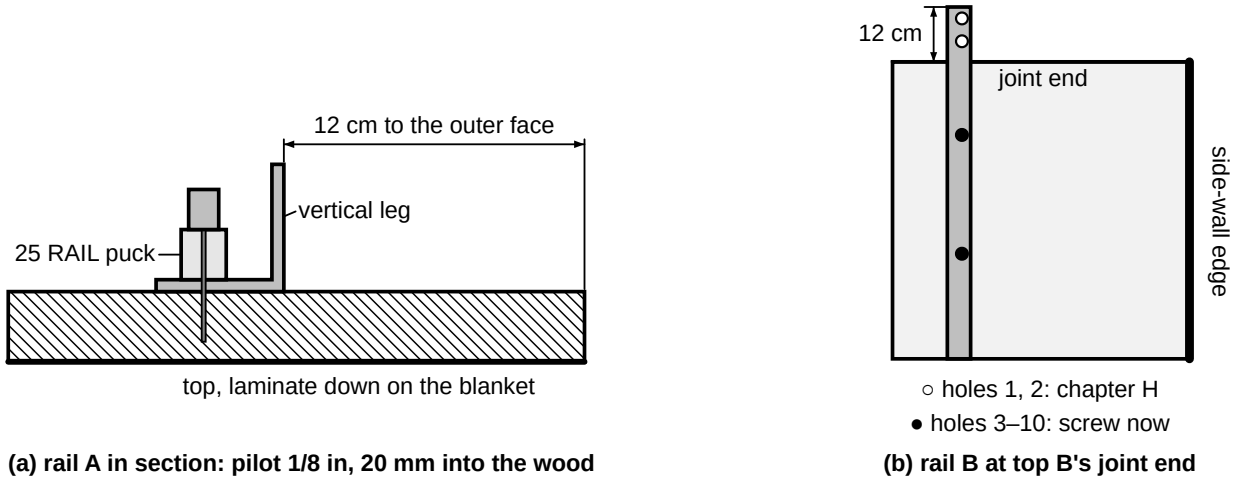


Fig. F-9 Rail A on top A, face down (cm). The rail lies on its pencil line from chapter D, flat leg against the top, wall end 1 cm in from the taped side-wall end. Check 12 cm from the front edge to the vertical leg's outer face at both ends and in the middle. Black: the two end screws, driven first.



(a) rail A in section: pilot 1/8 in, 20 mm into the wood

(b) rail B at top B's joint end

Fig. F-10 (a) Rail A in section, the top face down, so the vertical leg points up (mm, except the 12 cm). The 25 RAIL puck stands on the flat leg: the 1/8 in pilot goes 20 mm into the wood. (b) Rail B on top B, face down: its corner end projects 12 cm past the joint end; holes 1 and 2 (open) lie in the projection.

4 **Rail A.** Clamp it on its line, flat leg against the top, wall end at x = 1 cm. Measure 12 cm from the front edge to the vertical leg's outer face at both ends and in the middle. Through each rail hole drill a 1/8 in pilot with the **25 RAIL** puck standing on the flat leg (20 mm into the wood). Drive the two end screws first, re-check the 12 cm, then the rest: **H03** #10 × 1 in Robertson on the #2 bit in the 3 in extension (the chuck would hit the vertical leg otherwise), until each head seats. Nine screws.

5 **Rail B.** Clamp it on its line with the corner end projecting 12 cm past top B's joint end. Pilot and screw holes 3 to 10 only, the same way: eight screws. Holes 1 and 2 go into top A in chapter H.

The printed rail end caps go on later, once the desk stands; they are handed: cap A fits rail A's far end, cap B fits rail A's wall end and rail B's far end. Rail B's corner end is bolted to rail A and takes no cap.

F · Step 6: place the ALEX units

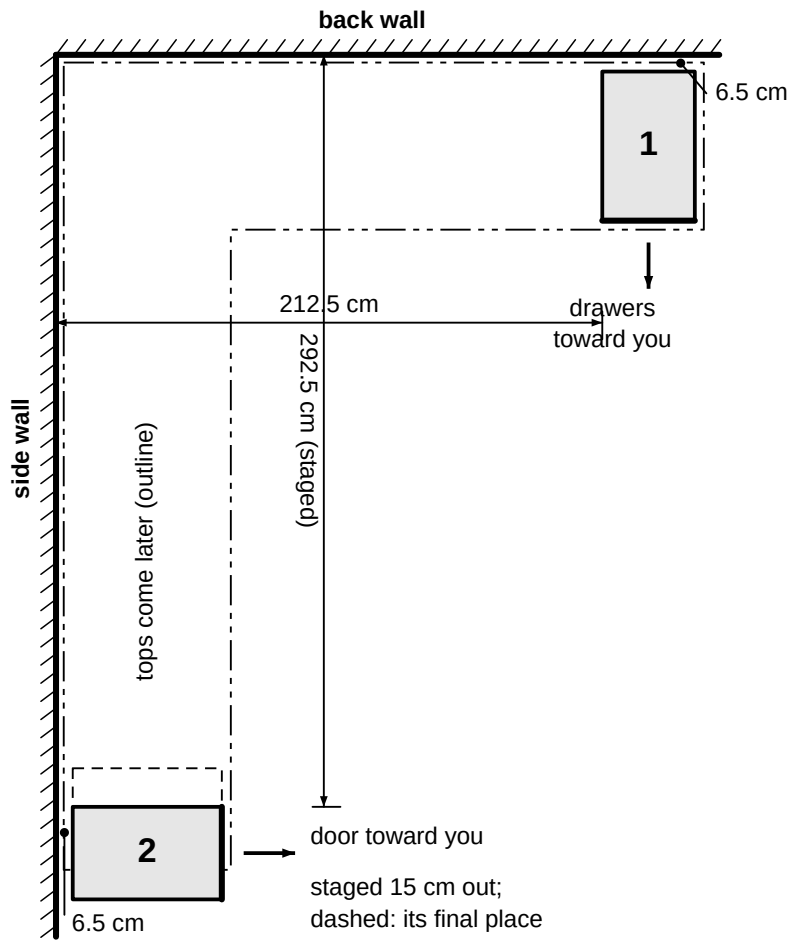


Fig. F-11 Plan from above. The tops are not up yet, so the cabinets are set from the walls (cm). Heavy line: the drawer fronts of ALEX 1 face you (+y); the door of ALEX 2 faces you (+x). ALEX 2 solid: where it stands now; dashed: its final place.

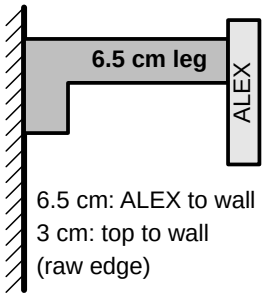


Fig. F-12 Wall gauge (P18): the 6.5 cm leg just fits between the wall and a cabinet; the 3 cm leg checks a top's raw edge.

6

Stand **ALEX 1** (P03) with its back 6.5 cm off the back wall and its near side 212.5 cm from the side wall, drawers toward you. Stand **ALEX 2** (P04) with its side 6.5 cm off the side wall and, for now, its near end **292.5 cm** from the back wall, door toward you: 15 cm short of its final 277.5 cm, so that top B can be stood on it in chapter G with rail B's 12 cm projection clear of top A. It is pushed home in chapter G. Painter's tape on the floor round each one saves guessing.

No rings yet: they are fitted in chapter G by tracing the pads. A 50 mm pad in a 52 mm ring has about 1 mm each side, closer than a cabinet can be placed with a tape. Once the tops are on, 3.5 cm of top shows all round each cabinet and the tops stand 3 cm off the walls.

Optional ALEX feet (decided at the start of this chapter): fit them now, before the cabinets stand. Unscrew the stock feet, lay the unit on its back, hook the corner template over each bottom corner and use its inner three holes (r 14.5 mm). Pilot 5/64 in, 12 mm deep with the 12 ALEX puck; three #6 × 5/8 flat heads per foot down the wells, four feet per unit, all the same height.

! **MEASURE ON HARDWARE** MEASURE ON HARDWARE

Before fitting ALEX feet, measure the ALEX bottom panel at an edge. 14 mm or more is fine: the 12 ALEX pilot is 12 mm deep and the #6 × 5/8 screw reaches only 9.9 mm into the panel through the foot's 6 mm roof, so 4 mm remains at 14 mm. Under 14 mm: do not fit them. The top panel measured 15.9 mm, so the bottom is likely similar and likely fine.

F · The ring, and the check (used in chapter G)

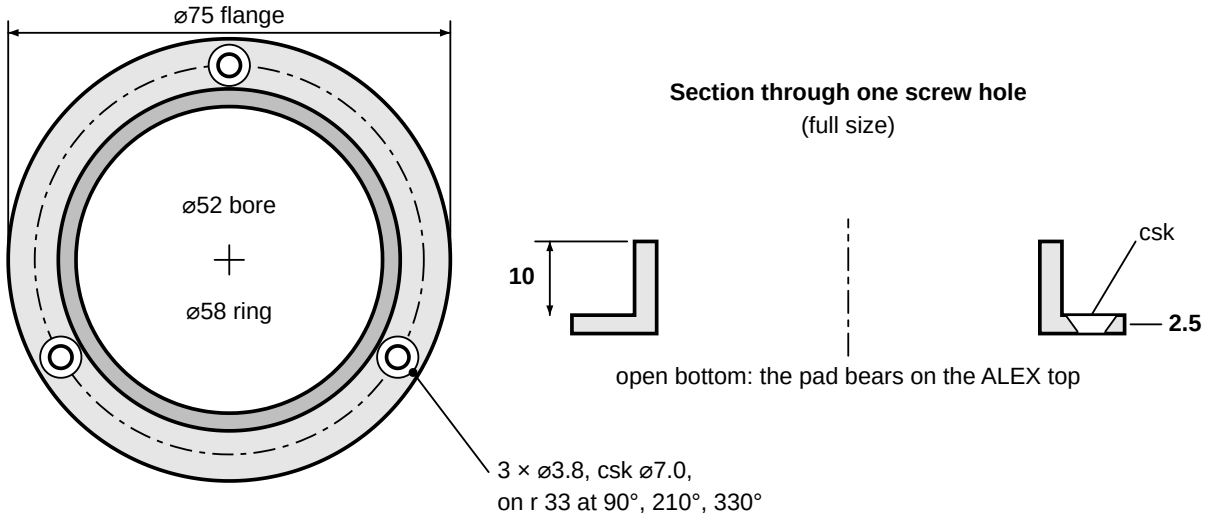


Fig. F-13 Printed locating ring (P10), from `parts/structure/foot-ring.scad`. **VERIFIED** from its own geometry file, mm: bore ø52, ring ø58 (3 wall) and 10 tall on a ø75 × 2.5 flange, three ø3.8 holes countersunk ø7.0 on r 33. Print flange-down, PETG or ABS.

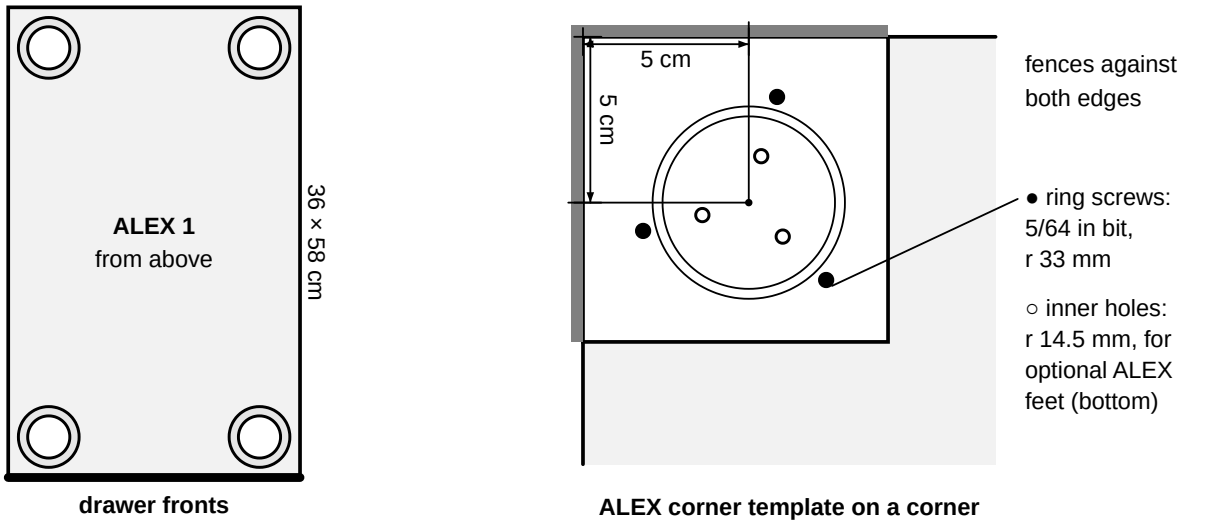


Fig. F-14 The check, after the rings are fitted in chapter G. Left: ALEX 1's top with its four rings, centres 5 cm in from each corner; ALEX 2 is the same at all four corners. Right: drop the ALEX corner template over a corner, fences against both edges: its ring outline should sit over the fitted ring. It is a check, not the way to place the rings.

The ring is round: when it is screwed down in chapter G it is turned until its three holes meet whatever three pilots it finds. The corner template is a check only. If a trace is more than 2 mm off the template's circle, the cabinet is not where this chapter placed it: fix the cabinet, not the ring.

F · Foot, pad and ring in section

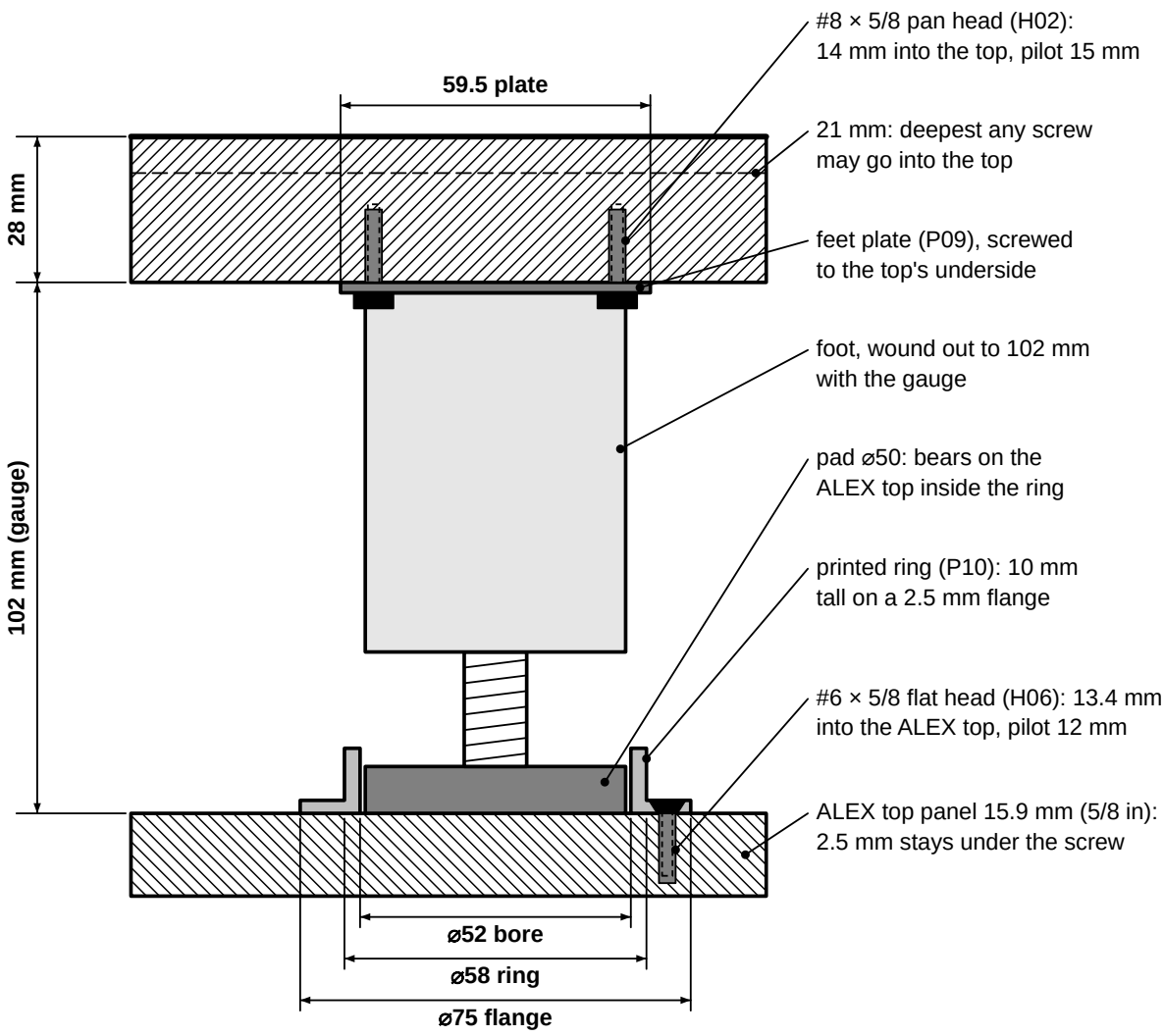


Fig. F-15 Section through one foot, its ring and one ring screw (mm). Hatched: the top above and the ALEX top panel below, both cut. The foot's internal shape is drawn schematically: only its 102 mm setting and ⌀50 mm body and pad are known. Heights above the floor: surface 83.0 cm, underside 80.2 cm, ALEX top 70 cm ASSUMPTION / FIELD CHECK.

Dimension	mm	Confidence
Top thickness	28	MEASURE ON HARDWARE expect 28 mm
Foot, plate face to pad (gauge)	102	SOURCE-DEFINED
Feet plate, hole spacing	59.5 / 46.8	VERIFIED
Pad diameter	50	MEASURE ON HARDWARE before relying on the 52 mm bore
Ring bore / ring / flange	52 / 58 / 75 × 2.5	VERIFIED geometry file
ALEX top panel thickness	15.9	VERIFIED 5/8 in, 23 Oct 2026; 2.5 mm under each screw

F · Load path

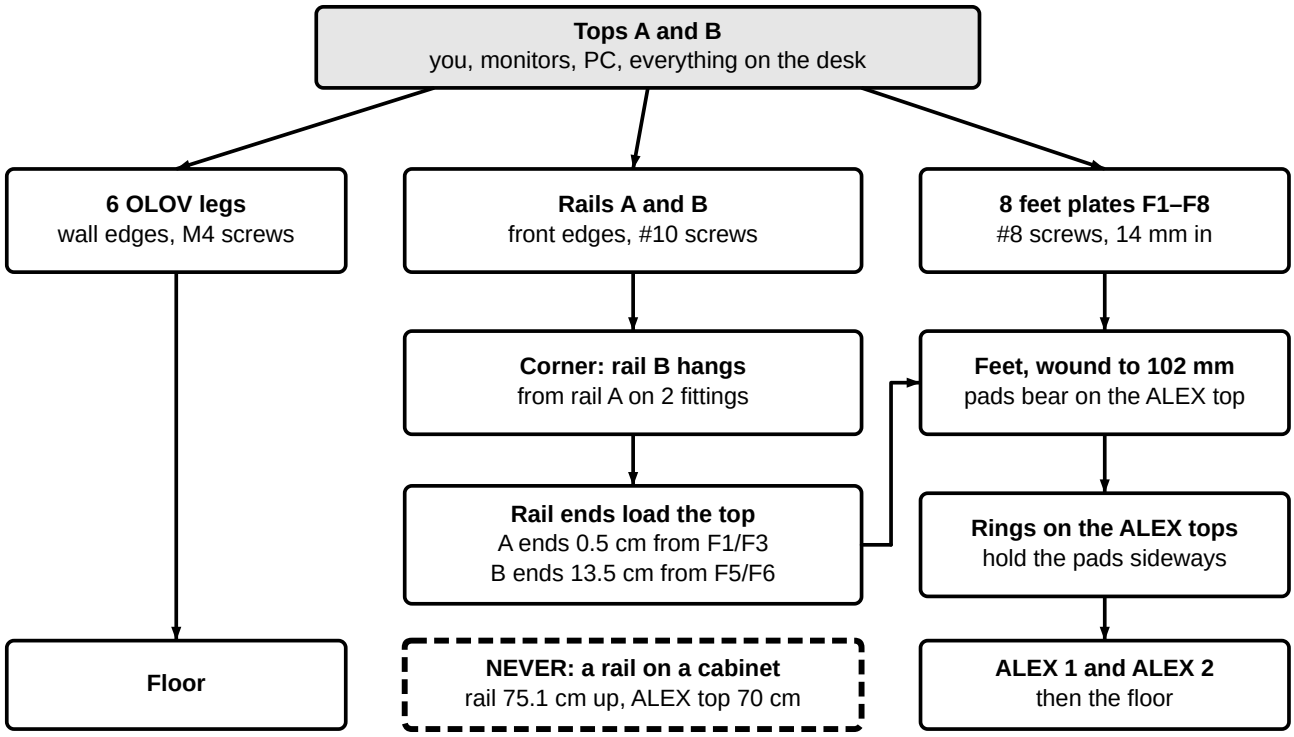
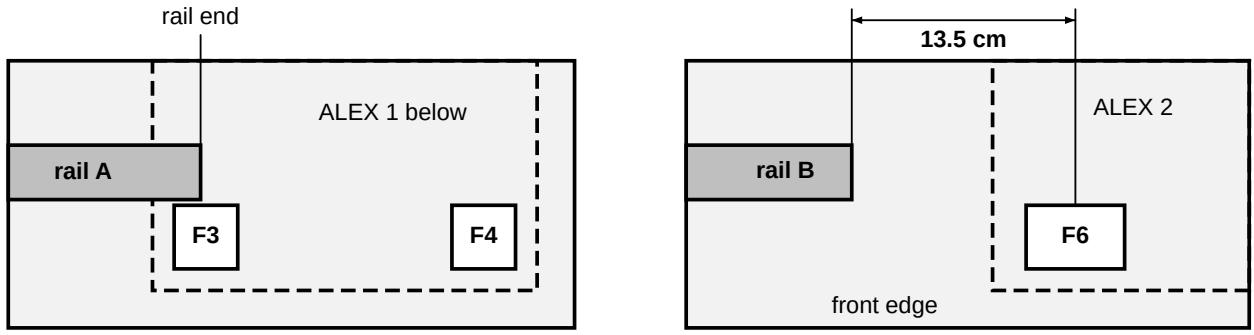


Fig. F-16 What carries what. The legs and the feet carry the tops to the floor. The rails carry the front edges; their ends load the top, and the top loads the nearest feet plates. The pads bear on the ALEX tops; the rings only stop them sliding.



(a) rail A's far end, plan
rail end x 214 cm
F1/F3 centres x 214.5 cm

(b) rail B's end, plan (y to the right)
rail end y 266 cm · ALEX 2 face y 274.5 cm
F5/F6 centres y 279.5 cm

Fig. F-17 Where the rails stop (cm from the datum; front edges at the bottom). (a) Rail A ends at x 214 cm, 0.5 cm short of the F1/F3 centres at 214.5 cm, beside F3's plate. (b) Rail B ends at y 266 cm; the top bridges the 13.5 cm to F5/F6.

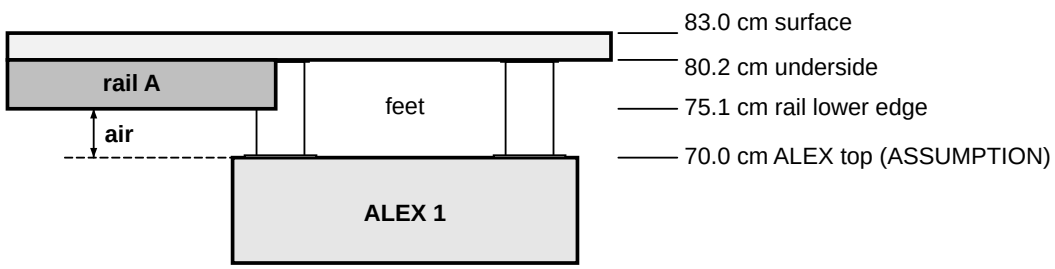


Fig. F-18 Elevation at ALEX 1, seen from the room. Rail A's end passes over the cabinet in plan but stays in the air above it: its lower edge is 75.1 cm above the floor, the ALEX top 70 cm.

F · The assembled underside

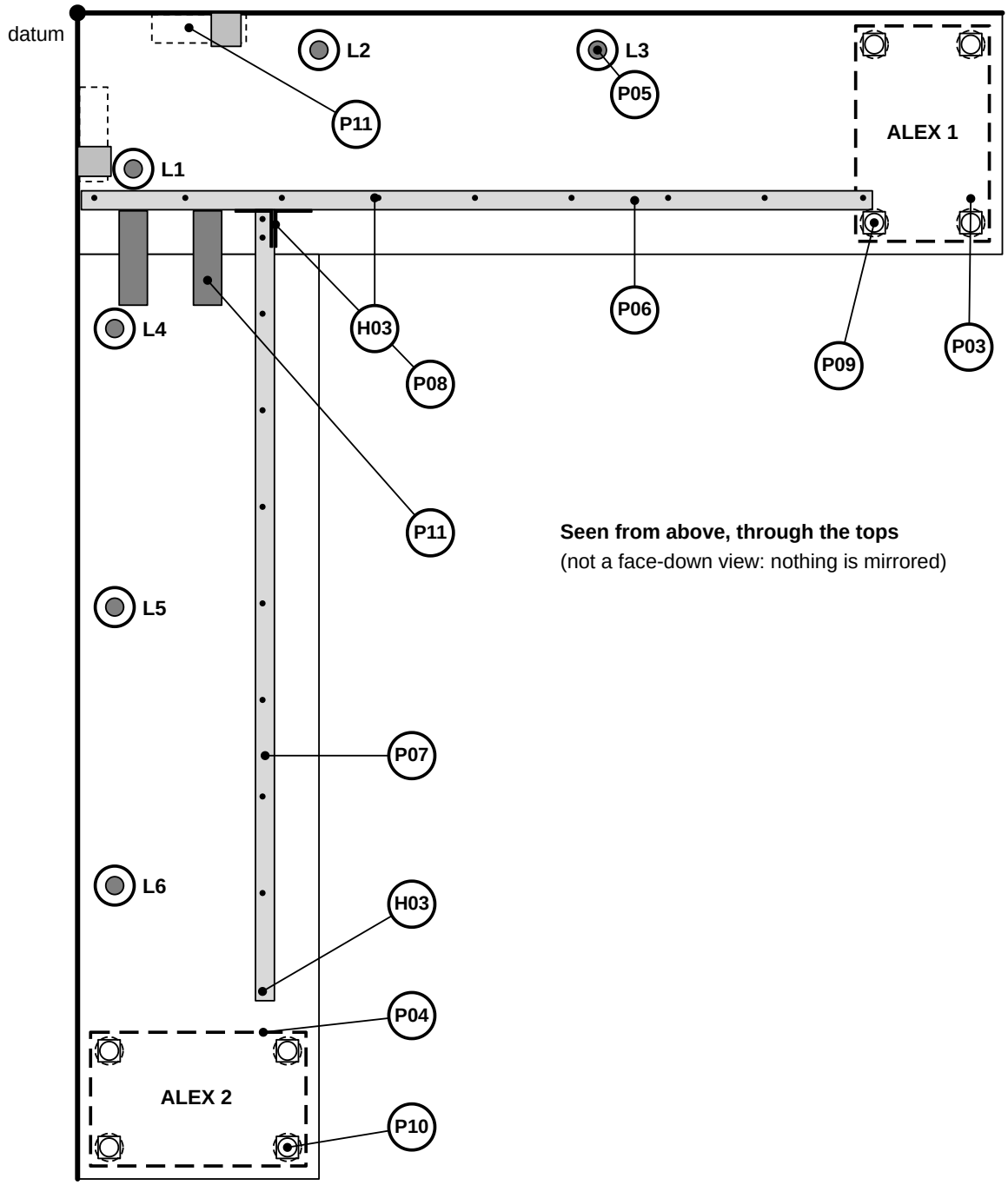


Fig. F-19 Everything under the tops once the desk is built, seen from above through the tops. Datum (0, 0) at top A's back-left corner. The rails go on in steps 4 and 5 of this chapter, the rings in chapter G, the corner fittings and joining plates in chapter H; all are shown here to check against.

No.	Part	No.	Part
P03	ALEX 1, drawer unit 36 × 58 × 70 cm	**P08**	Gusseted 90-degree strut fitting x 2
P04	ALEX 2, storage unit with door 36 × 58 × 70 cm	**P09**	Round adjustable foot 4 in x 8
P05	OLOV leg x 6, 105 mm plate	**P10**	Printed locating ring x 8
P06	Rail A, steel angle 213 cm	**P11**	Mending plate 10 x 3 x 0.12 in x 4 (two joining, two clamp pads)
P07	Rail B, steel angle 213 cm	**H03**	#10 x 1 in pan head, Robertson (rails into the tops), 19

Next: chapter G, standing the tops on the cabinets and fitting the rings.